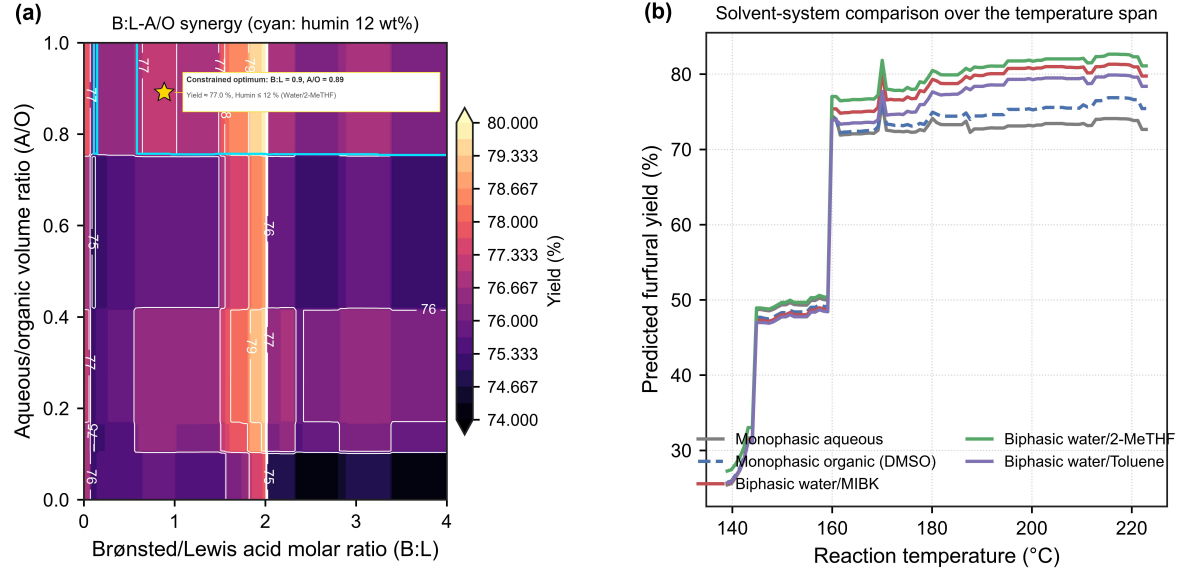

(a)
B:L-A/O synergy (cyan: humin 12 wt%)
Constrained optimum: B:L = 0.9, A/O = 0.89
Yield = 77.0 %, Humin ≤ 12 % (Water/2-MeTHF)
Aqueous/organic volume ratio (A/O)
Brønsted/Lewis acid molar ratio (B:L)
Yield (%)
80.000
79.333
78.667
78.000
77.333
76.667
76.000
75.333
74.667
74.000
(b)
Solvent-system comparison over the temperature span
Predicted furfural yield (%)
Reaction temperature (°C)
Monophasic aqueous
Monophasic organic (DMSO)
Biphasic water/MIBK
Biphasic water/2-MeTHF
Biphasic water/Toluene